Two Multiplicative Lemmas in Number Theory

The Four-Number Lemma and the Committee–Subcommittee Identity

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Overview

This note develops two factorisation principles that turn multiplicative equations into algebraic identities, then traces several consequences of each.

The **Four-Number Lemma** says that any equation of the form $ab = cd$ in positive integers can be rewritten in four parameters p, q, r, s, after which the original constraint becomes a tautology. Three problems — a sum of nth powers, a Diophantine equation $x^2+Dy^2 = z^2$, and a polynomial divisibility statement — yield to this single substitution.

The **Committee–Subcommittee Identity**

$$\binom{n}{k}\binom{k}{m} = \binom{n}{m}\binom{n-m}{k-m},$$

familiar from counting arguments, becomes a number-theoretic tool when both sides are read as positive integers: it forces non-trivial divisibility relations among binomial coefficients.

Part I. The Four-Number Lemma

1 The Lemma

Four-Number Lemma. *Let a, b, c, d be positive integers with $ab = cd$. Then there exist positive integers p, q, r, s with*

$$a = pq, \qquad b = rs, \qquad c = pr, \qquad d = qs.$$

Moreover, p, q, r, s can be chosen so that $\gcd(q, r) = 1$ *and* $\gcd(p, s) = 1$.

How to read this

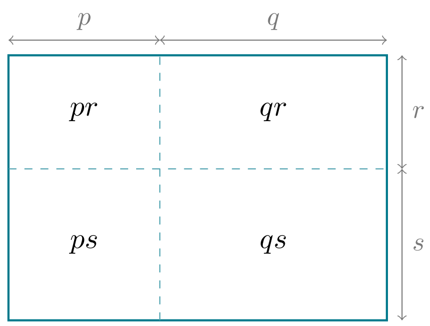
A multiplicative equation in four unknowns ($ab = cd$) is replaced by four parameters

(p, q, r, s). After substitution, the original equation becomes

$$(pq)(rs) = (pr)(qs),$$

which is automatic. So whenever a problem reduces to $ab = cd$, the lemma offers a substitution that converts the constraint into something we can manipulate algebraically: sums become products, divisibility becomes coprimality, and Diophantine equations split open.

Geometric picture



A rectangle of width $p+q$ and height $r+s$ has area $(p+q)(r+s)$. The dashed cuts split it into four sub-rectangles of areas pr, qr, ps, qs. The original equation $ab = cd$ says: this rectangle can be cut into two strips in two different ways — horizontal strips of areas a, b, or vertical strips of areas c, d. The four corner pieces are the *common refinement* of those two ways.

Proof. Set $p = \gcd(a, c)$ and write

$$a = pq, \qquad c = pr, \qquad \gcd(q, r) = 1.$$

Substituting into $ab = cd$:

$$(pq)b = (pr)d \quad \Longrightarrow \quad qb = rd.$$

Now $q \mid rd$ and $\gcd(q, r) = 1$, so by Euclid's lemma $q \mid d$. Write $d = qs$. Substituting back,

$$qb = r \cdot qs \quad \Longrightarrow \quad b = rs.$$

This gives the parameterisation. The condition $\gcd(p, s) = 1$ follows by extracting the global $\gcd(a, b, c, d)$ at the start — a normalisation step that does not affect the structure of the proof. □

2 The Fundamental GCD Identity

Theorem 1 (GCD identity). *If a, b, c, d are positive integers with $ab = cd$ and $\gcd(a, b, c, d) = 1$, then*

$$a = \gcd(a, c) \cdot \gcd(a, d).$$

Proof. By the lemma, $a = pq$, $c = pr$, $d = qs$, with $\gcd(q, r) = 1$ and $\gcd(p, s) = 1$. Then

$$\gcd(a, c) = \gcd(pq, pr) = p \gcd(q, r) = p,$$
$$\gcd(a, d) = \gcd(pq, qs) = q \gcd(p, s) = q.$$

Multiplying gives $\gcd(a, c) \cdot \gcd(a, d) = pq = a$. □

The point

a factorises canonically as a product of two *coprime* divisors — one shared with c, one shared with d. This decomposition is what the polynomial divisibility application below depends on.

3 Application 1: A Sum of nth Powers

Problem

Let a, b, c, d be positive integers with $ac = bd$. Prove that for every integer $n \geq 1$,

$$a^n + b^n + c^n + d^n$$

is composite.

The condition $ac = bd$ is the Four-Number Lemma equation in disguise: $a \cdot c = b \cdot d$. The lemma applies directly — but with a, c playing the role of one pair and b, d the other.

Solution. Apply the Four-Number Lemma to $ac = bd$:

$$a = pq, \quad c = rs, \quad b = pr, \quad d = qs.$$

Substitute into the sum:

$$\begin{aligned} a^n + b^n + c^n + d^n &= (pq)^n + (pr)^n + (rs)^n + (qs)^n \\ &= p^n q^n + p^n r^n + r^n s^n + q^n s^n \\ &= p^n(q^n + r^n) + s^n(q^n + r^n) \\ &= (p^n + s^n)(q^n + r^n). \end{aligned}$$

Each of the two factors is at least $1^n + 1^n = 2$, so the sum is a product of two integers each ≥ 2, hence composite. □

Remark. The factorisation $p^n(q^n + r^n) + s^n(q^n + r^n) = (p^n + s^n)(q^n + r^n)$ is the engine: the lemma's parameterisation is exactly designed so that this grouping works. Notice how the pair $(q^n + r^n)$ is shared between two of the four terms — this is no coincidence, it is the diagonal structure of the rectangle picture.

4 Application 2: The Equation $x^2 + 6y^2 = z^2$

Problem

Find a parametric family of positive integer solutions to

$$x^2 + 6y^2 = z^2.$$

The equation rearranges to

$$z^2 - x^2 = 6y^2 \iff (z-x)(z+x) = 6y^2,$$

which is the multiplicative equation we need. The factor $6 = 2 \cdot 3$ is what generates a family: different ways of distributing the factor 6 across the parameters give different solution families.

Solution. Rearrange: $(z-x)(z+x) = 6y \cdot y$. Apply the Four-Number Lemma:

$$z - x = pq, \quad z + x = rs, \quad 6y = pr, \quad y = qs.$$

The third and fourth equations give $pr = 6qs$. Split the factor 6 as $p = 2k$ and $r = 3m$, so the equation becomes $6km = 6qs$, i.e. $qs = km$. This is itself a multiplicative equation, and a second application of the lemma yields

$$q = ab, \quad s = cd, \quad k = ac, \quad m = bd.$$

Reassembling,

$$p = 2k = 2ac, \quad q = ab, \quad r = 3m = 3bd, \quad s = cd,$$

so

$$z - x = pq = 2a^2bc, \qquad z + x = rs = 3bcd^2.$$

Adding and subtracting,

$$2z = bc(2a^2 + 3d^2), \qquad 2x = bc(3d^2 - 2a^2).$$

Absorbing the common factor $bc/2$ (or bc when an even factor is present) into a single parameter K, and renaming $u = d$, $v = a$, gives the family

$$\boxed{x = K\,|3u^2 - 2v^2|, \qquad y = 2Kuv, \qquad z = K\,(3u^2 + 2v^2).}$$

A direct check confirms $x^2 + 6y^2 = z^2$ for all positive integers K, u, v with $3u^2 \neq 2v^2$. $\square$

Generalisation: $x^2 + Dy^2 = z^2$

Theorem 2 (General weighted Pythagorean family). *For any factorisation $D = d_1 d_2$ into positive integers, the equation $x^2 + Dy^2 = z^2$ admits the family of positive integer solutions*

$$x = K\,|d_1u^2 - d_2v^2|, \qquad y = 2Kuv, \qquad z = K\,(d_1u^2 + d_2v^2),$$

where K, u, v are positive integers with $d_1u^2 \neq d_2v^2$. Each factorisation of D produces a distinct family.

Why this works. The proof of Application 2 used only the factorisation $6 = 2 \cdot 3$. Replacing the constants $2, 3$ by general d_1, d_2 and tracking the algebra gives the formula above. The case $D = 1$ with $d_1 = d_2 = 1$ recovers the classical Pythagorean parameterisation

$$(u^2 - v^2,\ 2uv,\ u^2 + v^2).$$

Why each factorisation matters

For $D = 12$, the splittings $1 \cdot 12$, $2 \cdot 6$, $3 \cdot 4$ each generate a genuinely different solution family. The Four-Number Lemma is the engine that produces all of them.

5 Application 3: A Polynomial Divisibility Statement

Problem

Suppose a, b, c, d are positive integers with $ab = cd$ and $\gcd(a, b, c, d) = 1$. Let $P(x)$ and $Q(x)$ be polynomials with integer coefficients, and assume

$$a \mid P(c)\,Q(d).$$

Prove that

$$\gcd(a, c) \mid P(0)\,Q(d) \qquad \text{and} \qquad \gcd(a, d) \mid P(c)\,Q(0).$$

The bridge is Theorem 1: under the coprimality assumption,

$$a = \gcd(a, c) \cdot \gcd(a, d),$$

where the two factors are themselves coprime. We split the divisibility relation across these two coprime factors and check each one separately.

Solution. Set $g_1 = \gcd(a, c)$ and $g_2 = \gcd(a, d)$. By Theorem 1, $a = g_1 g_2$ with $\gcd(g_1, g_2) = 1$. Since $g_1 \mid c$ and $g_2 \mid d$, we have

$$c \equiv 0 \pmod{g_1}, \qquad d \equiv 0 \pmod{g_2}.$$

For the first conclusion, work modulo g_1. Because P has integer coefficients, every term of $P(c) - P(0)$ contains a factor of c, so

$$c \equiv 0 \pmod{g_1} \quad \Longrightarrow \quad P(c) \equiv P(0) \pmod{g_1}.$$

Multiplying by $Q(d)$,

$$P(c)\,Q(d) \equiv P(0)\,Q(d) \pmod{g_1}.$$

By hypothesis $a \mid P(c)\,Q(d)$, and since $g_1 \mid a$, we have $g_1 \mid P(c)\,Q(d)$. Combining,

$$P(0)\,Q(d) \equiv P(c)\,Q(d) \equiv 0 \pmod{g_1},$$

so $g_1 \mid P(0)\,Q(d)$, as required.

The second conclusion is symmetric. Using $d \equiv 0 \pmod{g_2}$ and the same polynomial identity for Q,

$$P(c)\,Q(d) \equiv P(c)\,Q(0) \pmod{g_2}.$$

Since $g_2 \mid a \mid P(c)\,Q(d)$, we conclude $g_2 \mid P(c)\,Q(0)$. $\square$

The mechanism

The polynomial identity $P(x) - P(0) = x \cdot \tilde{P}(x)$ for some integer polynomial $\tilde{P}$ tells us that $P(c) \equiv P(0)$ modulo any divisor of c. The GCD identity supplies precisely the right divisors: g_1 is the part of a shared with c (so $P(c)$ collapses to $P(0)$), and g_2 is the part shared with d (so $Q(d)$ collapses to $Q(0)$). Each part of a "sees" only one of the two polynomials.

6 Application 4: A Cyclic Fraction Sum

Problem

Let a, b, c, d be *pairwise distinct* positive integers, and suppose

$$S = \frac{a}{a+b} + \frac{b}{b+c} + \frac{c}{c+d} + \frac{d}{d+a}$$

is an integer. Prove that $a^2 + b^2 + c^2 + d^2$ is composite.

Solution. We first show $S = 2$, then deduce $ac = bd$ from this, and finally apply the Four-Number Lemma to factor the sum of squares.

For the lower bound on S, replace each denominator by the larger common quantity $a+b+c+d$:

$$S > \frac{a}{a+b+c+d} + \frac{b}{a+b+c+d} + \frac{c}{a+b+c+d} + \frac{d}{a+b+c+d} = 1.$$

For the upper bound, write $\dfrac{a}{a+b} = 1 - \dfrac{b}{a+b}$ in each term:

$$S = 4 - \left(\frac{b}{a+b} + \frac{c}{b+c} + \frac{d}{c+d} + \frac{a}{d+a} \right).$$

The bracket is the cyclic shift of the original sum, and the same lower-bound argument shows it exceeds 1, so $S < 4 - 1 = 3$. Combining, $1 < S < 3$, and integrality forces

$$S = 2.$$

Group the four fractions into alternating pairs and combine each pair over a common denominator:

$$\left(\frac{a}{a+b} + \frac{c}{c+d} \right) + \left(\frac{b}{b+c} + \frac{d}{d+a} \right) = 2.$$

Subtracting 1 from each pair and simplifying gives

$$\frac{ac - bd}{(a+b)(c+d)} + \frac{bd - ac}{(b+c)(d+a)} = 0,$$

which factors as

$$(ac - bd) \left(\frac{1}{(a+b)(c+d)} - \frac{1}{(b+c)(d+a)} \right) = 0. \tag{1}$$

The bracket vanishes iff $(a+b)(c+d) = (b+c)(d+a)$. Expanding and cancelling the common terms ac and bd leaves $ad + bc = ab + cd$, which rearranges to

$$(a-c)(d-b) = 0.$$

Since a, b, c, d are pairwise distinct, both $a \neq c$ and $b \neq d$, so the bracket is nonzero. Therefore equation (1) forces

$$ac = bd.$$

By the Four-Number Lemma applied to $ac = bd$, there exist positive integers p, q, r, s with

$$a = pq, \quad c = rs, \quad b = pr, \quad d = qs.$$

Substituting into the sum of squares,

$$\begin{aligned} a^2 + b^2 + c^2 + d^2 &= (pq)^2 + (pr)^2 + (rs)^2 + (qs)^2 \\ &= p^2(q^2 + r^2) + s^2(q^2 + r^2) \\ &= (p^2 + s^2)(q^2 + r^2). \end{aligned}$$

Each factor is at least $1^2 + 1^2 = 2$, so $a^2 + b^2 + c^2 + d^2$ is composite. □

A worked example

Take $p = 1$, $q = 2$, $r = 3$, $s = 4$. Then

$$a = pq = 2, \quad b = pr = 3, \quad c = rs = 12, \quad d = qs = 8,$$

which are pairwise distinct, and $ac = 24 = bd$. We verify S:

$$S = \tfrac{2}{5} + \tfrac{3}{15} + \tfrac{12}{20} + \tfrac{8}{10} = \tfrac{2}{5} + \tfrac{1}{5} + \tfrac{3}{5} + \tfrac{4}{5} = 2. \checkmark$$

The sum of squares is $4 + 9 + 144 + 64 = 221$, and the predicted factorisation matches:

$$(p^2 + s^2)(q^2 + r^2) = (1 + 16)(4 + 9) = 17 \cdot 13 = 221. \checkmark$$

The big picture

A bound on a fraction sum forces an integer value; that value forces a multiplicative relation; the lemma turns the relation into an algebraic factorisation. Three different techniques, one final answer.

Summary of Part I

- **The Four-Number Lemma.** $ab = cd$ implies $a = pq$, $b = rs$, $c = pr$, $d = qs$ with $\gcd(q, r) = \gcd(p, s) = 1$.
- **Geometric reading.** The lemma is the multiplicative analogue of the common refinement of two partitions of a rectangle.
- **GCD identity.** If $\gcd(a, b, c, d) = 1$, then $a = \gcd(a, c) \cdot \gcd(a, d)$, with the two factors coprime.

- **Three classes of application:**
  - Sums of nth powers factor as $(p^n + s^n)(q^n + r^n)$, hence are composite.
  - Diophantine equations $x^2 + Dy^2 = z^2$ have one parametric family per factorisation $D = d_1 d_2$.
  - Polynomial divisibility "splits" across coprime factors of a, via the GCD identity.

Part II. The Committee–Subcommittee Identity

7 The Identity, Read Multiplicatively

Committee–Subcommittee Identity. *For non-negative integers $n \geq k \geq m$,*

$$\binom{n}{k}\binom{k}{m} = \binom{n}{m}\binom{n-m}{k-m}.$$

The combinatorial proof is the standard one: both sides count the number of ways to choose, from n people, a committee of k and within it a subcommittee of m — the LHS picks committee first, the RHS picks subcommittee first. We have used this identity before for evaluating sums and integrals.

A change of perspective

> Both sides of the identity are positive integers. So the equation does not just compute a count: it forces *divisibility* relations among the binomial coefficients on each side. Each side is a multiple of every binomial coefficient appearing in it. Exploiting this is the entire point of Part II.

This perspective gives us two theorems and one celebrated corollary.

8 Theorem 1: Pairwise Common Factors in Pascal's Triangle

Theorem 3. *For any integer $n > 2$ and any distinct k, m with $0 < k, m < n$,*

$$\gcd\left(\binom{n}{k}, \binom{n}{m}\right) > 1.$$

In words: *no two interior entries of row n of Pascal's triangle (for $n > 2$) are coprime.* Each row has many entries built from the same factorial $n!$, yet the internal arithmetic is constrained enough that any two of them must share a prime factor.

Proof. Without loss of generality assume $k > m$. The committee–subcommittee identity reads

$$\binom{n}{k}\binom{k}{m} = \binom{n}{m}\binom{n-m}{k-m}.$$

The left side is divisible by $\binom{n}{k}$, hence so is the right side. That is, $\binom{n}{k}$ divides $\binom{n}{m}\binom{n-m}{k-m}$.

Suppose for contradiction that $\gcd\left(\binom{n}{k}, \binom{n}{m}\right) = 1$. Then by Euclid's lemma, $\binom{n}{k}$ must divide the second factor:

$$\binom{n}{k} \,\Big|\, \binom{n-m}{k-m}.$$

But this is impossible. Since $0 < m$, we have shrunk the range of selection by a positive amount, so

$$\binom{n-m}{k-m} < \binom{n}{k}$$

(intuitively: fewer choices available, fewer ways to choose). A positive integer cannot be divided by a strictly larger positive integer.

This contradiction shows $\gcd\left(\binom{n}{k}, \binom{n}{m}\right) > 1$. $\square$

Remark. The strict inequality $\binom{n-m}{k-m} < \binom{n}{k}$ deserves a moment's thought. Multiplicatively,

$$\frac{\binom{n}{k}}{\binom{n-m}{k-m}} = \frac{n!\,(n-k)!\,(k-m)!}{k!\,(n-k)!\,(n-m)!} \cdot \frac{1}{1} = \frac{n \cdot (n-1) \cdots (n-m+1)}{k \cdot (k-1) \cdots (k-m+1)} > 1,$$

since each factor on top is strictly larger than the corresponding factor on the bottom (as $n > k$).

9 Theorem 2: A Refined Divisibility Lemma

Theorem 4. *For positive integers n, k with $1 \le k \le n$,*

$$\frac{n}{\gcd(n,k)} \,\Big|\, \binom{n}{k}.$$

This refines the basic fact that $\binom{n}{k}$ is an integer. The refinement weights the divisor by how much n and k share. When $\gcd(n,k) = 1$, the theorem reads

$$n \,\Big|\, \binom{n}{k},$$

which is exactly what we need for the corollary below.

Proof. Set $m = 1$ in the committee–subcommittee identity:

$$\binom{n}{k}\binom{k}{1} = \binom{n}{1}\binom{n-1}{k-1}, \quad \text{i.e.} \quad k\binom{n}{k} = n\binom{n-1}{k-1}.$$

Let $d = \gcd(n,k)$, and write

$$n = dn', \quad k = dk', \quad \gcd(n', k') = 1.$$

The identity becomes

$$dk'\binom{n}{k} = dn'\binom{n-1}{k-1} \implies k'\binom{n}{k} = n'\binom{n-1}{k-1}.$$

So n' divides $k'\binom{n}{k}$. Since $\gcd(n', k') = 1$, by Euclid's lemma $n' \mid \binom{n}{k}$.
Substituting back $n' = n/\gcd(n, k)$:

$$\frac{n}{\gcd(n,k)} \,\Big|\, \binom{n}{k}.$$

□

10 Corollary: Prime Power Divisibility

Corollary 1. *Let p be a prime and $a \geq 1$. For every k with $0 < k < p^a$,*

$$p \,\Big|\, \binom{p^a}{k}.$$

The case $a = 1$ recovers the well-known fact that $p \mid \binom{p}{k}$ for $0 < k < p$ — the key ingredient in proving the binomial theorem modulo p and, ultimately, the Frobenius endomorphism $x \mapsto x^p$.

Proof. By Theorem 4,

$$\frac{p^a}{\gcd(p^a,k)} \,\Big|\, \binom{p^a}{k}.$$

Since $0 < k < p^a$, the gcd $\gcd(p^a, k)$ is a power of p strictly less than p^a. Write

$$\gcd(p^a, k) = p^b, \quad 0 \leq b < a.$$

Then

$$\frac{p^a}{p^b} = p^{a-b} \,\Big|\, \binom{p^a}{k},$$

and $a - b \geq 1$, so $p \mid \binom{p^a}{k}$. □

Why this matters

The corollary $p \mid \binom{p^a}{k}$ is the arithmetic backbone of *Lucas's theorem*, the binomial theorem in characteristic p, and the Frobenius endomorphism in commutative algebra. The proof shown here — via Theorem 2 — is one of the cleanest derivations available, and it all comes from setting $m = 1$ in a familiar combinatorial identity.

Summary of Part II

- Reading $\binom{n}{k}\binom{k}{m} = \binom{n}{m}\binom{n-m}{k-m}$ multiplicatively forces divisibility relations among binomial coefficients.

- **Theorem 1.** No two interior entries of row n of Pascal's triangle ($n > 2$) are coprime.
- **Theorem 2.** $\dfrac{n}{\gcd(n,k)} \,\Big|\, \binom{n}{k}$ for all $1 \le k \le n$.
- **Corollary.** $p \mid \binom{p^a}{k}$ for all primes p and $0 < k < p^a$.

Practice Problems

Problem 1

Find all positive integer solutions to $a^2 + b^2 = c^2 + d^2$ subject to $ab = cd$.
Hint. Apply the Four-Number Lemma to $ab = cd$ directly.

Problem 2

Show that for any prime p and any $1 \le k \le p - 1$,

$$\binom{p}{k} \equiv 0 \pmod{p}.$$

Use this to prove Fermat's little theorem: $a^p \equiv a \pmod{p}$ for all integers a.

Problem 3

Compute

$$\gcd\left(\binom{10}{1}, \binom{10}{2}, \ldots, \binom{10}{9}\right)$$

explicitly, and conjecture a general formula in terms of n.